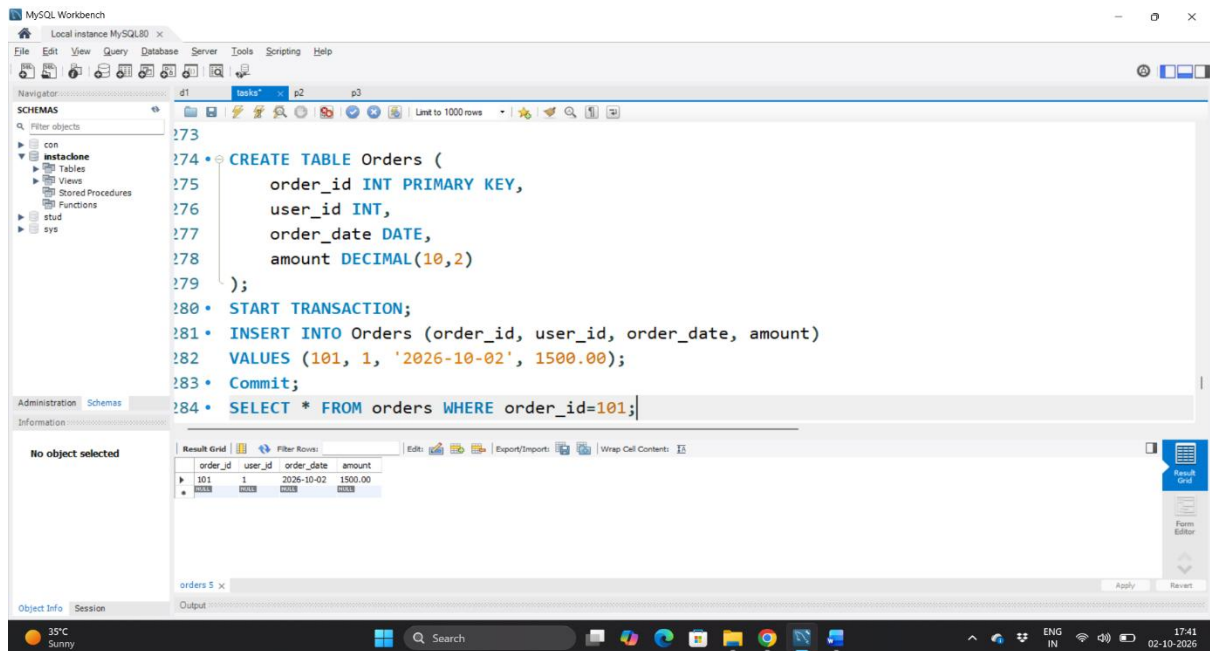
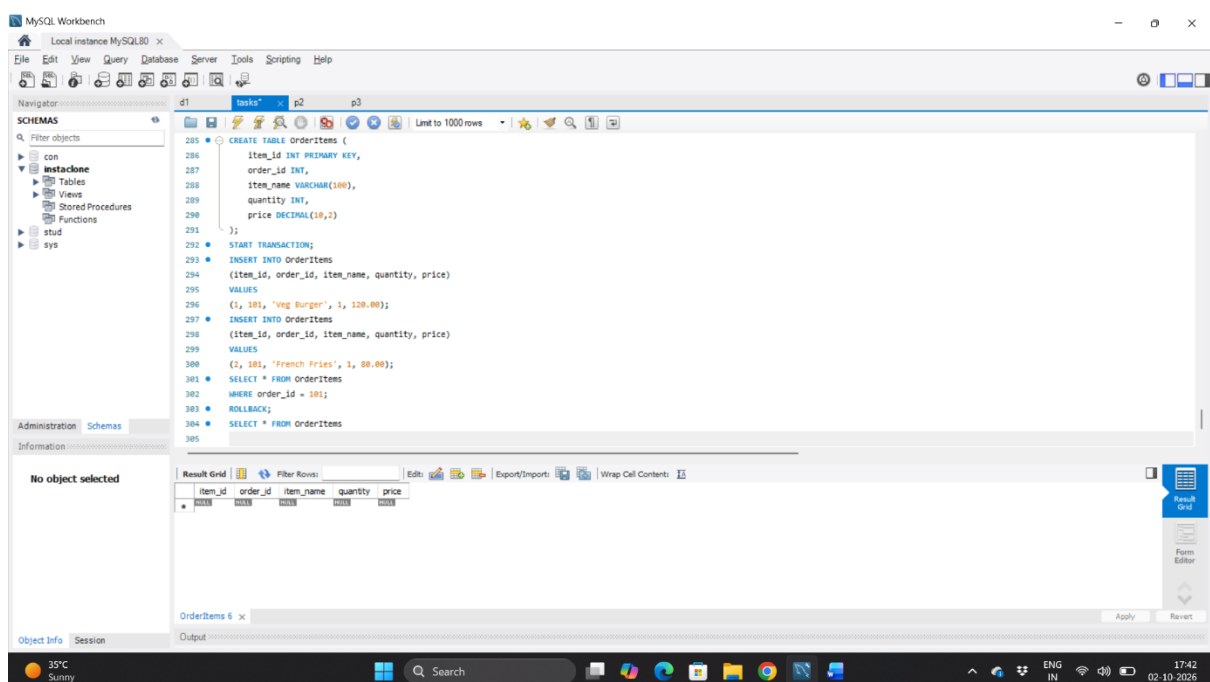


Session 9

Q1. Create a SQL script to insert a new order into an Orders table, then use COMMIT to save the transaction and verify that the new order persists after reconnecting to the database.



Q2. Simulate a Zomato-style food order process: insert two new items into an OrderItems table, then use ROLLBACK to undo the changes before committing. Check that no new items remain in the table after rollback.



Q3. In a Flipkart-like shopping cart scenario, use SAVEPOINT to mark a point after adding a product to the Cart table, add another product, then use ROLLBACK TO SAVEPOINT to undo only the last addition. Show the final contents of the Cart table.

The screenshot shows the MySQL Workbench interface with a script editor containing the following SQL code:

```

314 START TRANSACTION;
315 INSERT INTO Cart
316 (cart_id, user_id, product_name, quantity, price)
317 VALUES
318 (1, 101, 'Laptop', 1, 50000.00);
319 SAVEPOINT product_added;
320 INSERT INTO Cart
321 (cart_id, user_id, product_name, quantity, price)
322 VALUES
323 (2, 101, 'Wireless Mouse', 1, 800.00);
324 ROLLBACK TO SAVEPOINT product_added;
325 SELECT * FROM Cart;
326 COMMIT;

```

The Result Grid shows the final state of the Cart table:

cart_id	user_id	product_name	quantity	price
1	101	Laptop	1	50000.00

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
23	17:47:15	SAVEPOINT product_added	0 row(s) affected	0.000 sec
24	17:47:24	INSERT INTO Cart (cart_id, user_id, product_name, quantity, price) VALUES (2, 101, 'Wireless Mouse', 1, 800.00)	1 row(s) affected	0.000 sec
25	17:47:35	ROLLBACK TO SAVEPOINT product_added	0 row(s) affected	0.000 sec
26	17:47:43	SELECT * FROM Cart LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
27	17:47:57	COMMIT	0 row(s) affected	0.047 sec
28	17:48:01	SELECT * FROM Cart LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Q4. Write a trigger on a Wallet table that automatically deducts the purchase amount from the user's balance whenever a new transaction is inserted, similar to how Paytm updates wallet balance after a payment.

The screenshot shows the MySQL Workbench interface with the following SQL code in the script editor:

```

328 CREATE TABLE Wallet (
329     user_id INT PRIMARY KEY,
330     balance DECIMAL(10,2)
331 );
332
333 CREATE TABLE Transactions (
334     transaction_id INT PRIMARY KEY,
335     user_id INT,
336     purchase_amount DECIMAL(10,2),
337     FOREIGN KEY (user_id) REFERENCES Wallet(user_id)
338 );

```

The Result Grid shows the initial state of the Wallet table:

user_id	balance
1	2000.00

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
32	17:55:14	CREATE TABLE Wallet (user_id INT PRIMARY KEY, balance DECIMAL(10,2))	0 row(s) affected	0.031 sec
33	17:55:23	CREATE TABLE Transactions (transaction_id INT PRIMARY KEY, user_id INT, purchase_amount DE...	0 row(s) affected	0.078 sec
34	17:55:49	CREATE TRIGGER deduct_wallet_balance AFTER INSERT ON Transactions FOR EACH ROW BEGIN U...	0 row(s) affected	0.016 sec
35	17:56:22	INSERT INTO Wallet (user_id, balance) VALUES (1, 2000.00)	1 row(s) affected	0.016 sec
36	17:56:32	INSERT INTO Transactions (transaction_id, user_id, purchase_amount) VALUES (101, 1, 1200.00)	1 row(s) affected	0.016 sec
37	17:56:55	SELECT * FROM Wallet LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

con

instancione

Tables

Views

Stored Procedures

Functions

stud

sys

Administration Schemas

Information

No object selected

Wallet 9

Output

Action Output

Time Action Message Duration / Fetch

32 17:55:14 CREATE TABLE Wallet (user_id INT PRIMARY KEY, balance DECIMAL(10,2)) 0 row(s) affected 0.031 sec

33 17:55:23 CREATE TABLE Transactions (transaction_id INT PRIMARY KEY, user_id INT, purchase_amount DE... 0 row(s) affected 0.078 sec

34 17:55:49 CREATE TRIGGER deduct_wallet_balance AFTER INSERT ON Transactions FOR EACH ROW BEGIN U... 0 row(s) affected 0.016 sec

35 17:56:22 INSERT INTO Wallet (user_id, balance) VALUES (1, 5000.00) 1 row(s) affected 0.016 sec

36 17:56:32 INSERT INTO Transactions (transaction_id, user_id, purchase_amount) VALUES (101, 1, 1200.00) 1 row(s) affected 0.016 sec

37 17:56:55 SELECT * FROM Wallet LIMIT 0, 1000 1 row(s) returned 0.015 sec / 0.000 sec

350

340 DELIMITER //

341

342 CREATE TRIGGER deduct_wallet_balance

343 AFTER INSERT ON Transactions

344 FOR EACH ROW

345 BEGIN

346 UPDATE Wallet

347 SET balance = balance - NEW.purchase_amount

348 WHERE user_id = NEW.user_id;

349 END //

350

Result Grid

Filter Rows:

Edits Export/Import Wrap Cell Contents

user_id balance

1 3800.00

35°C Sunny

Search

ENG IN

17:58 02-10-2026

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

con

instancione

Tables

Views

Stored Procedures

Functions

stud

sys

Administration Schemas

Information

No object selected

Wallet 9

Output

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Time Action Message Duration / Fetch

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351 DELIMITER ;

352

353 INSERT INTO Wallet (user_id, balance)

354 VALUES (1, 5000.00);

355

356 INSERT INTO Transactions (transaction_id, user_id, purchase_amount)

357 VALUES (101, 1, 1200.00);

358

359 SELECT * FROM Wallet;

360

361

Result Grid

Filter Rows:

Edits Export/Import Wrap Cell Contents

user_id balance

1 3800.00

35°C Sunny

Search

ENG IN

17:58 02-10-2026

Q5. Use ChatGPT to generate a SQL test scenario where a trigger incorrectly updates a user's balance after an expense entry. Paste the scenario and your corrected trigger code, explaining how you fixed the error. Hint: Ask ChatGPT for a buggy trigger example and a test case that exposes the bug.

